- write answer of Which of the following is NOT a primary activity in system engineering? a) Requirement analysis b) Concept design c) Financial analysis d) System integration and testing
- The system architecture defines: a) The hardware and software components b) The relationships and interactions between components c) Both a and b d) None of the above
- The iterative development process in system engineering typically involves: a) Defining all requirements upfront b) Developing the system in a linear, step-by-step manner c) Refining requirements and design through repeated cycles d) Focusing primarily on coding and testing
- Which of the following is NOT a typical stakeholder in a system engineering project? a) Users b) Developers c) Regulators d) Suppliers
- Risk management in system engineering involves: a) Identifying potential risks b) Assessing the likelihood and impact of risks c) Developing mitigation strategies d) All of the above
- Which of the following is NOT a characteristic of a good system requirements specification? a) Clear and unambiguous b) Complete and consistent c) Testable and verifiable d) Open to interpretation
- Functional requirements define: a) What the system must do b) How the system will do it c) Both a and b d) None of the above
- Non-functional requirements define: a) Performance, quality, and usability attributes b) The specific functionalities of the system c) The hardware and software components d) The development process
- System trade-off analysis involves: a) Evaluating the pros and cons of different design options b) Selecting the cheapest option c) Following the engineer's intuition d) Ignoring trade-offs altogether
- Integration testing in system engineering verifies: a) Individual components functionality b) How components interact with each other c) Both a and b d) System compliance with regulations
- System acceptance testing is typically performed by: a) The system engineers b) The developers c) The end users d) A combination of a and c
- Configuration management in system engineering ensures: a) Tracking and controlling changes to the system b) Efficient deployment and maintenance c) Both a and b d) User satisfaction
- Which of the following is NOT a benefit of using model-based system engineering? a) Improved communication and understanding b) Early identification of design flaws c) Increased development costs d) Enhanced system maintainability
- Open-loop systems: a) Have no feedback mechanism b) React automatically to changes in the environment c) Require manual intervention to control d) Both a and c
- Closed-loop systems: a) Utilize feedback to adjust their behavior b) Are independent of external influences c) Cannot adapt to changing conditions d) None of the above
- Emergent properties in systems arise from: a) The individual components of the system b) The interactions and relationships between components c) Both a and b d) External factors outside the system
- System of systems (SoS) engineering focuses on: a) Integrating independent systems to achieve a common goal b) Developing individual systems within a larger context c) Optimizing individual system performance d)
- Managing complex systems with many components Which of the following is NOT a typical SoS challenge? a) Decentralized decision-making b) Interoperability among disparate systems c) Standardized interfaces and protocols d) Centralized control and governance

- Agile system engineering practices emphasize: a) Iterative development with frequent feedback b) Detailed upfront planning and documentation c) Waterfall development with sequential phases d) Large teams and hierarchical structure

Answers

- Which of the following is NOT a primary activity in system engineering? Answer: c) Financial analysis
- The system architecture defines: Answer: c) Both a and b
- The iterative development process in system engineering typically involves: Answer: c) Refining requirements and design through repeated cycles
- Which of the following is NOT a typical stakeholder in a system engineering project? Answer: d) Suppliers
- Risk management in system engineering involves: Answer: d) All of the above
- Which of the following is NOT a characteristic of a good system requirements specification? Answer: d) Open to interpretation
- Functional requirements define: Answer: a) What the system must do
- Non-functional requirements define: Answer: a) Performance, quality, and usability attributes
- System trade-off analysis involves: Answer: a) Evaluating the pros and cons of different design options
- Integration testing in system engineering verifies: Answer: c) Both a and b
- System acceptance testing is typically performed by: Answer: c) The end users
- Configuration management in system engineering ensures: Answer: c) Both a and b
- Which of the following is NOT a benefit of using model-based system engineering? Answer: c) Increased development costs
- Open-loop systems: Answer: a) Have no feedback mechanism
- Closed-loop systems: Answer: a) Utilize feedback to adjust their behavior
- Emergent properties in systems arise from: Answer: c) Both a and b
- System of systems (SoS) engineering focuses on: Answer: a) Integrating independent systems to achieve a common goal
- Which of the following is NOT a typical SoS challenge? Answer: d) Centralized control and governance
- Agile system engineering practices emphasize: Answer: a) Iterative development with frequent feedback